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NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9648/02

Paper 2 Core Paper **15 September 2014**

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This document consists of **18** printed pages and **0** blank pages.

[Turn over

Section A

Answer **all** the questions in this section.

1 Lactose intolerance in humans is the inability to hydrolyse lactose due to the lack of the enzyme lactase in the alimentary canal. As a result, bacteria in the large intestines feed on the lactose and produces fatty acids and methane which lead to diarrhoea and flatulence.

Bacteria have been used to produce lactase on an industrial scale as a dietary supplement for people who are lactose intolerant. Human lactase consists of a single 160 kDa polypeptide chain that localizes to the brush border membrane of intestinal epithelial cells.

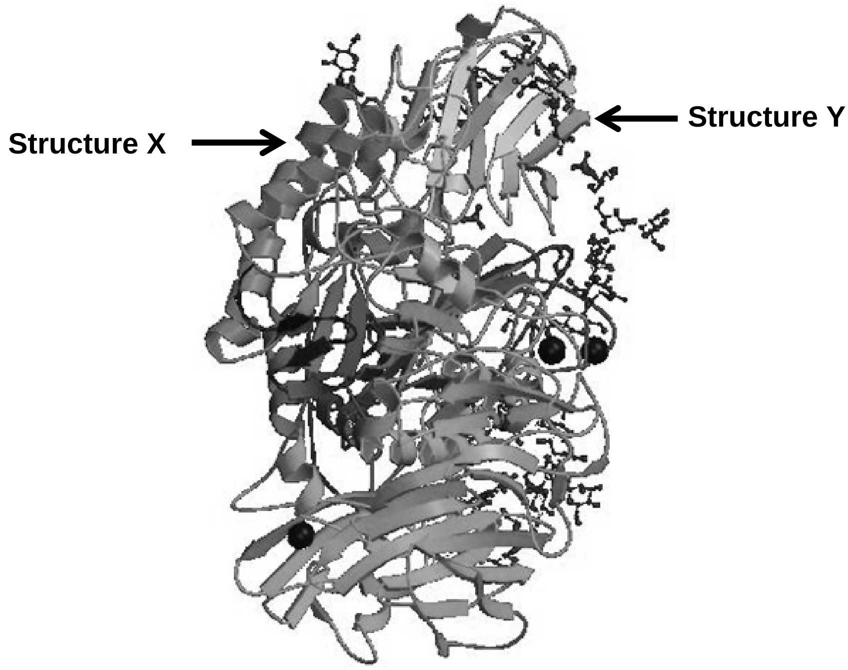

Fig. 1.1

(a) With reference to the **Fig. 1.1**,

(i) state the levels of organization seen in the structure of lactase.

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.... [1]

(ii) Describe structures **X** and **Y**

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.... [2]

(b) **Fig. 1.2** shows the hydrolysis of lactose.

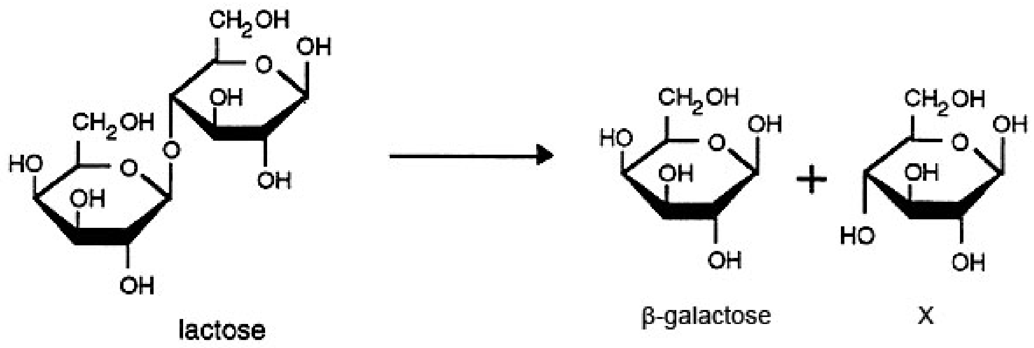

Fig. 1.2

(i) Describe the hydrolysis of lactose, naming the bond that is broken and product **X**.

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[2]

(c) Both lactase and lactose are protein and carbohydrate polymers respectively. Explain why there are fewer types of carbohydrate polymers compared to protein polymers.

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[2]

Cell organelles can be separated by centrifuging a cell extract in a sucrose density gradient. The organelles settle at the level in the sucrose solution which has the same density as their own.

The cells used to synthesize lactase were lysed and the cell extract centrifuged in a sucrose density gradient. Three distinct fractions of nuclei, mitochondria and ribosomes (in no particular order) were obtained. The three fractions **A, B** and **C** are shown in the **Fig. 1.3**.

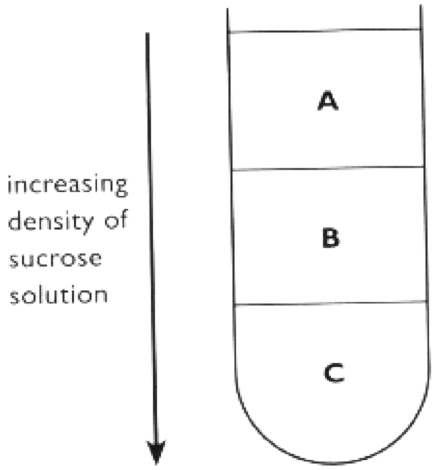

Fig. 1.3.

(d) Identify the organelles in each fraction and describe its role in the synthesis of lactase.

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[4]

[Total: 11]

2 The main function of a chromosome is to act as a ***template*** for the synthesis of RNA molecules, since only in this way does the genetic information stored in chromosomes become directly useful to the cell. RNA synthesis is a highly selective process. In most mammalian cells, only about 1% of the genetic information is transcribed into functional RNA sequences.

Before a cell can divide, it must produce a new copy of each of its chromosomes. Thus cell division is preceded by a special 'DNA-synthesis phase', during which each chromosome is replicated to produce two chromatids. These remain joined together at the centromere until the mitosis that soon follows.

(a) Explain what is meant by the term *template* (line 1).

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.... [3]

(b) Suggest why only about 1% of genetic information is transcribed into functional RNA sequences in most mammalian cells.

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.... [2]

(c) Outline how messenger RNA is synthesized in a eukaryotic cell.

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.... [3]

The graphs in **Fig.2.1** show the amount of DNA in the nuclei of cells taken from different parts of a mammalian testis during the formation of sperm cells. The formation of sperm cells involves mitosis and meiosis.

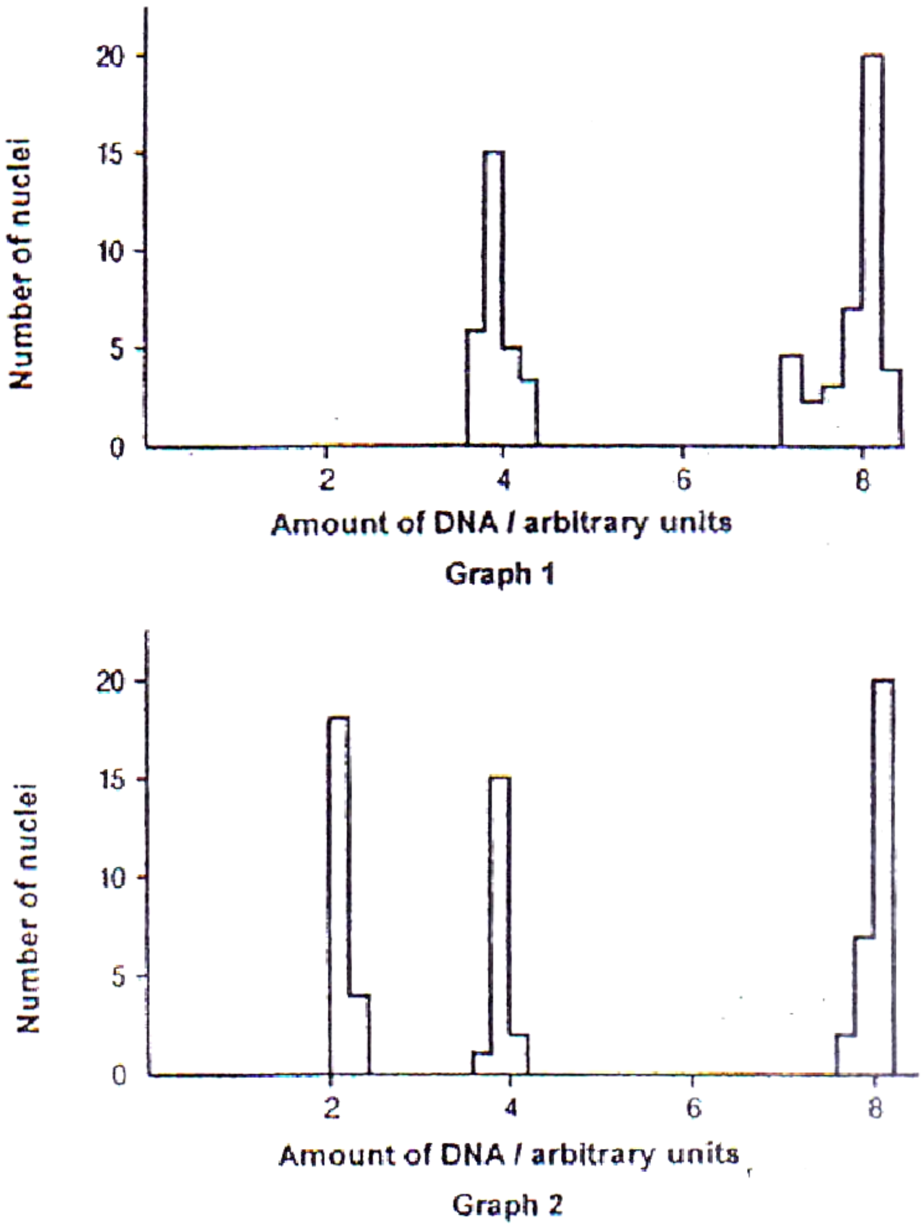

Fig. 2.1

(d) With reference to **Fig. 2.1**,

(i) explain the evidence which shows that Graph 1 represents cells undergoing mitosis.

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[2]

(ii) explain how the events of meiosis result in three groups of nuclei in Graph 2.

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[2]

[Total: 12]

3 The Ebola outbreak in West Africa is the world's deadliest to date and the World Health Organization (WHO) has now declared an international health emergency in response. It was first reported in February and according to the UN, 1013 people have died in Guinea, Liberia, Sierra Leone and Nigeria. (Why Ebola is so dangerous, 12 August 2014, BBC News)

Fig. 3.1 below shows some of the main structural features of the Ebola virus.

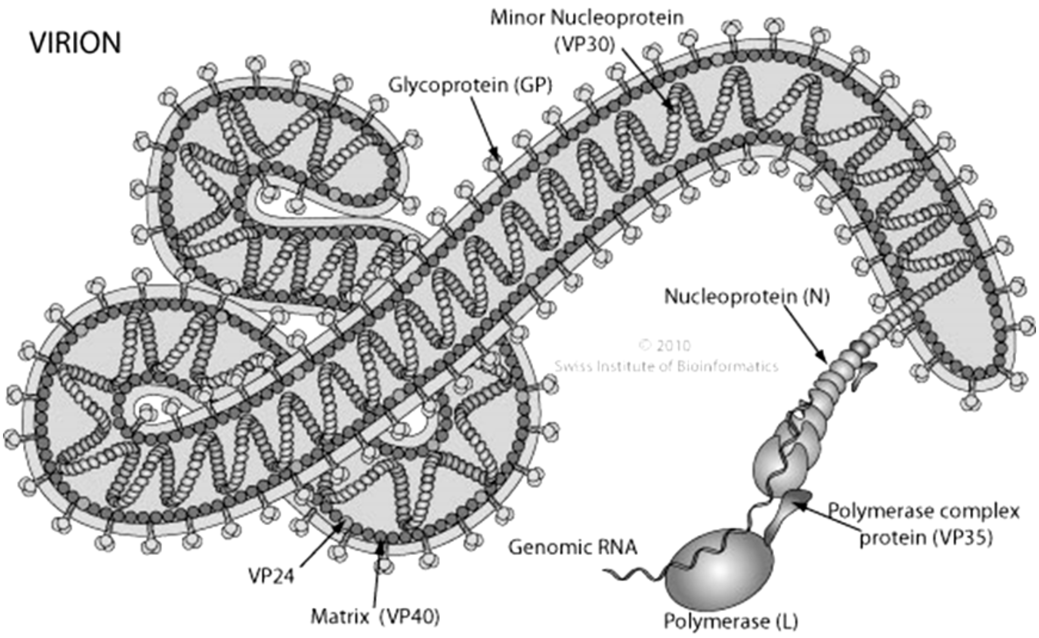

Fig. 3.1

The Ebola virus (EBOV) is similar to the influenza virus in many ways. EBOV infects the host cell in the same way as influenza does.

(a) Using your knowledge of influenza, suggest how EBOV may infect a host cell.

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.... [3]

The genome of EBOV contains one molecule of linear, negative sense RNA which is 19kb long. The genome encodes seven proteins, which include: nucleoprotein (NP), VP35, VP40, glycoprotein (GP), VP30, VP24, and L protein, which provides RNA-dependent RNA activity (Leroy *et al.*2001).

(b) The EBOV genome encodes L protein. Explain why such a gene is needed by the virus.

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[3]

(c) State how, despite having no ribosomes, EBOV is able to trigger the synthesis of its viral proteins.

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[1]

An experimental drug known as ZMapp has been approved by WHO to be used on three Ebola patients.

(d) Suggest how antiviral drugs can be used to halt the reproductive cycle of EBOV.

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[2]

(e) Even though Ebola is such a virulent disease, suggest why there is little progress in drug development.

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[1]

[Total: 10]

4 **Fig. 4.1** below shows the processes involved in gene expression in a mammalian cell.

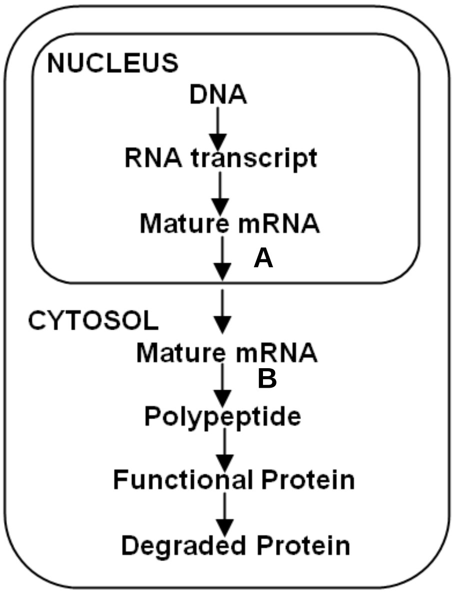

Fig. 4.1

(a) **(i)** Regulation of gene expression at A occurs in eukaryotes. State why it is absent in prokaryotes.

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[1]

(ii) Briefly describe one regulatory mechanism that can occur at process B.

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[1]

The Hayflick Limit is the number of times a normal human cell population will divide until cell division stops. Evidences have shown that telomeres associated with each cell's DNA will get slightly shorter with each new cell division until they shorten to a critical length. This is due to the end-replication problem that eukaryotic chromosomes have.

(b) Outline the end-replication problem in eukaryotic chromosomes.

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[3]

(c) Describe the role of telomeres in an eukaryotic cell.

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.... [2]

(d) Telomerase activity is essential in single celled organisms, while in humans and other vertebrates, it is only active in reproductive cells. Explain why regeneration of telomeres is not normally required in human somatic cells.

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.... [1]

(e) Telomerase activity is generally reactivated in tumour cells. Explain how this knowledge could be employed in the fight against cancer.

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.... [2]

[Total: 10]

5 The coat colours of Labrador Retriever dogs are determined by two pairs of unlinked genes B/b and E/e. The genotype of black coated dogs is B-E-, brown bbE- and golden --ee (where – indicates the presence of either the dominant or recessive allele).

(a) If heterozygous black (BbEe) dogs are mated, state the phenotypic ratio that can be expected in the progeny.

Show how you derive your answer.

[4]

The pedigree shown in **Fig. 5.1** below illustrates the inheritance of coat colour in Labrador Retriever, with black coats are represent be solid symbol, those with brown coats by dotted, with golden coats by open symbols.

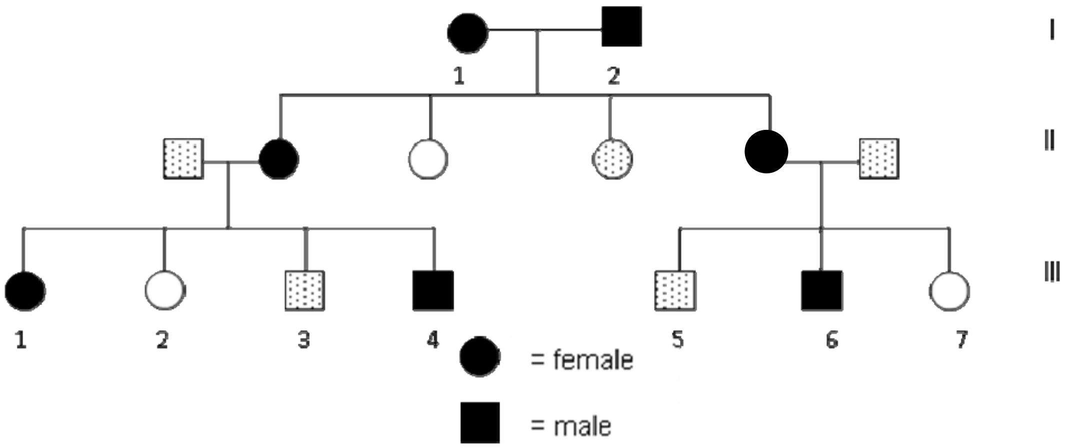

Fig. 5.1

(b) Suggest a molecular basis for the interaction described above.

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.... [3]

(c) State the genotypes of the parents I-1 and I-2. Give a reason for your answer.

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.... [2]

(d) State and explain the possible genotypes of III-4.

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.... [3]

[Total: 12]

6 **Fig. 6.1** below shows part of two nerve fibres and a synapse. The figures indicate the value in mV of the potential across the membrane between the cytoplasm of the fibres and the extracellular fluid at intervals along each fibre.

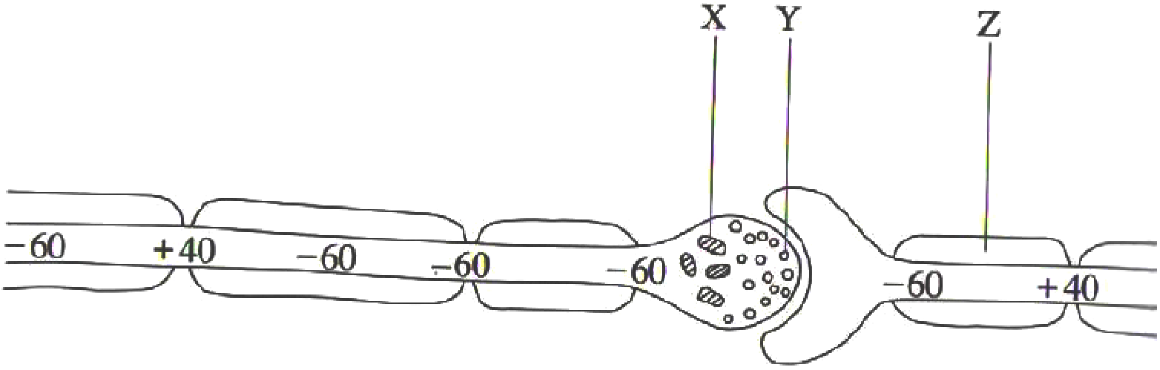

Fig. 6.1

(a) **(i)** Draw a circle around one region of the diagram where an action potential exists. Explain your choice.

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.... [2]

(ii) Explain the role of structures X and Y in the transmission of nerve impulse.

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.... [4]

(b) **Figure 6.2a** shows a normal nerve-muscle junction.

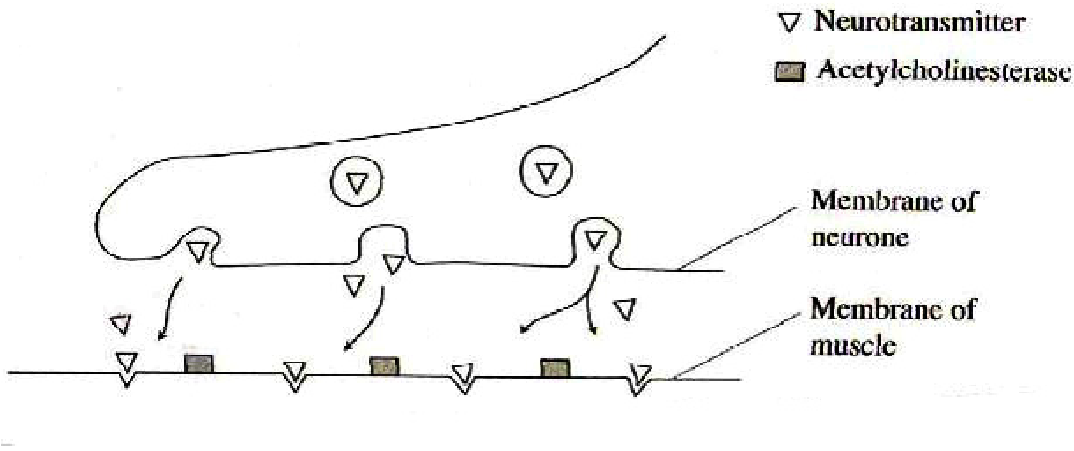

Fig. 6.2a

Using the evidence from **Fig. 6.2a** and **Fig. 6.2b**, briefly describe and explain the effect of muscle activity of **(i)** organophosphates and **(ii)** botulin toxin.

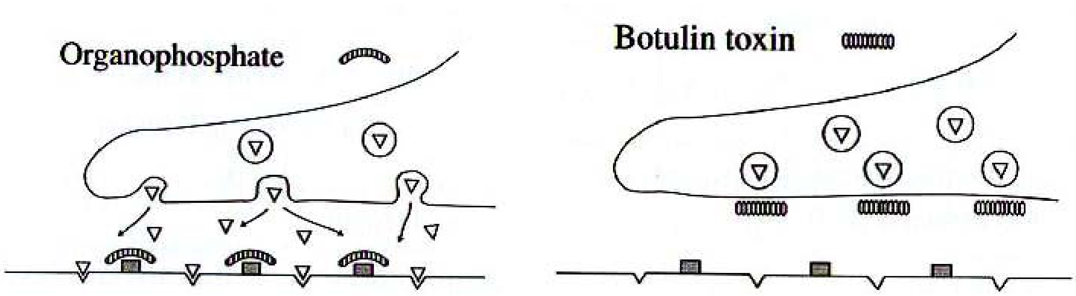

Fig. 6.2b

(i) organophosphate:

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.... [2]

(ii) botulin toxin:

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.... [2]

(c) Suggest the benefit of one neurone synapsing with more than one postsynaptic neurone.

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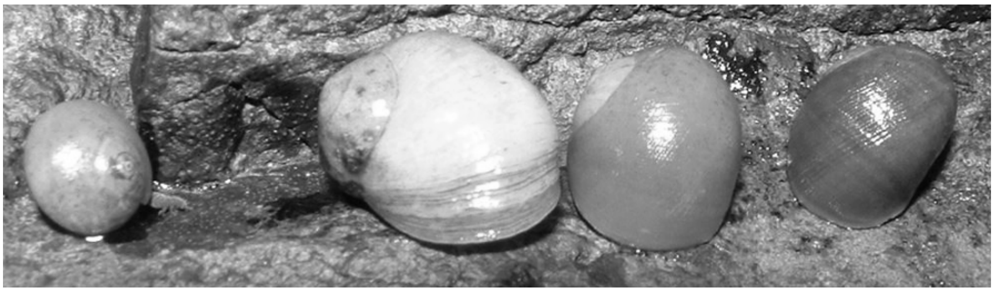
.... [1]

[Total: 11]

7 Read the passage carefully and then answer the questions that follow.



The flat periwinkles, *Littorina obtusata* and *Littorina mariae*, are two closely related marine snails that live on rocky sea shores around the coasts of the British Isles. On sheltered shores, *L. obtusata* is found mainly on the mid-shore on the seaweed *Ascophyllum nodosum* on which it feeds. *L. mariae* occurs on the lower shore on the seaweed *Fucus serratus*, feeding on the microscopic plants living on the surface of the seaweed.

The two species probably diverged from a common ancestor and they now differ markedly in both the size and colour of their shells. *L. obtusata* live for 3-4 years, breeding more than once. Their shells are larger (15-17mm) and usually green. *L. mariae* live for one year and breed only once. Their shells are smaller (9-12mm) and are usually yellow. *L. mariae* living lower on the shore is susceptible to greater predation pressure than *L. obtusata* on the mid-shore. This is due to the larger numbers of crabs which have longer foraging times on the lower shore compared to the mid-shore because of the tidal cycle.

On the lower shore, natural selection favours the annual life cycle of *L. mariae* with early maturity, rapid reproduction and short lifespan. On the mid-shore, *L. obtusata* has slow development, delayed reproduction and relatively long life-span. This species is prevented from extending its zone downwards by predations on the lower shore.

(a) Explain why there are no *L. mariae* larger than 12mm found on the shore.

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[2]

(b) Suggest why shell colour may be significant in avoiding predation.

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[2]

(c) Describe the role of natural selection in maintaining *L. mariae* as a separate species from *L. obtusata*.

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.... [4]

(d) Suggest a reason for the absence of

(i) *L. mariae* from the mid-shore

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(ii) *L. obtusata* from the lower shore

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.... [2]

(e) Flat periwinkles are widely distributed in all parts of the world.
Explain why different species of periwinkles found in different parts of the world can evolve similar traits.

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.... [2]

(f) Suggest how researchers can determine the evolutionary relationship of two periwinkle populations.

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.... [2]

[Total: 14]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8 **(a)** Contrast the structures and reproductive cycles of lambda phage and human immunodeficiency virus (HIV). [7]

(b) Explain how control elements influence transcription. [6]

(c) Describe the processes and significances, with relevant examples, of gene amplification in eukaryotes. [7]

9 **(a)** Describe the differences between continuous and discontinuous variation. [5]

(b) Explain with examples, how embryological, anatomical and molecular homologies support Darwin's theory of evolution. [8]

(c) Explain, with appropriate examples, how recessive alleles may be preserved in a natural population. [7]

[Total: 20]